

EE168 – Lab 2 Report

Layout Design, DRC, LVS, NAND, Parasitics, Ring Oscillator

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Summary

This lab walked through the full CMOS layout flow: drawing and verifying an inverter, extending the same checks to NAND and a hierarchical ring oscillator, and closing the loop with DRC/LVS. We built simple testbenches, captured DC and transient waveforms, and compared pre- and post-layout behavior to see parasitic impact. The ring oscillator exercise reinforced device sizing, hierarchy, and clean routing, while the repeated DRC/LVS runs highlighted how small spacing or connectivity errors surface in the tools. Overall, I gained confidence in taking schematics to layout, validating them with Calibre-style checks, and interpreting waveforms to confirm functional intent.

1 Inverter Layout, DRC, LVS

1.1 Inverter Layout

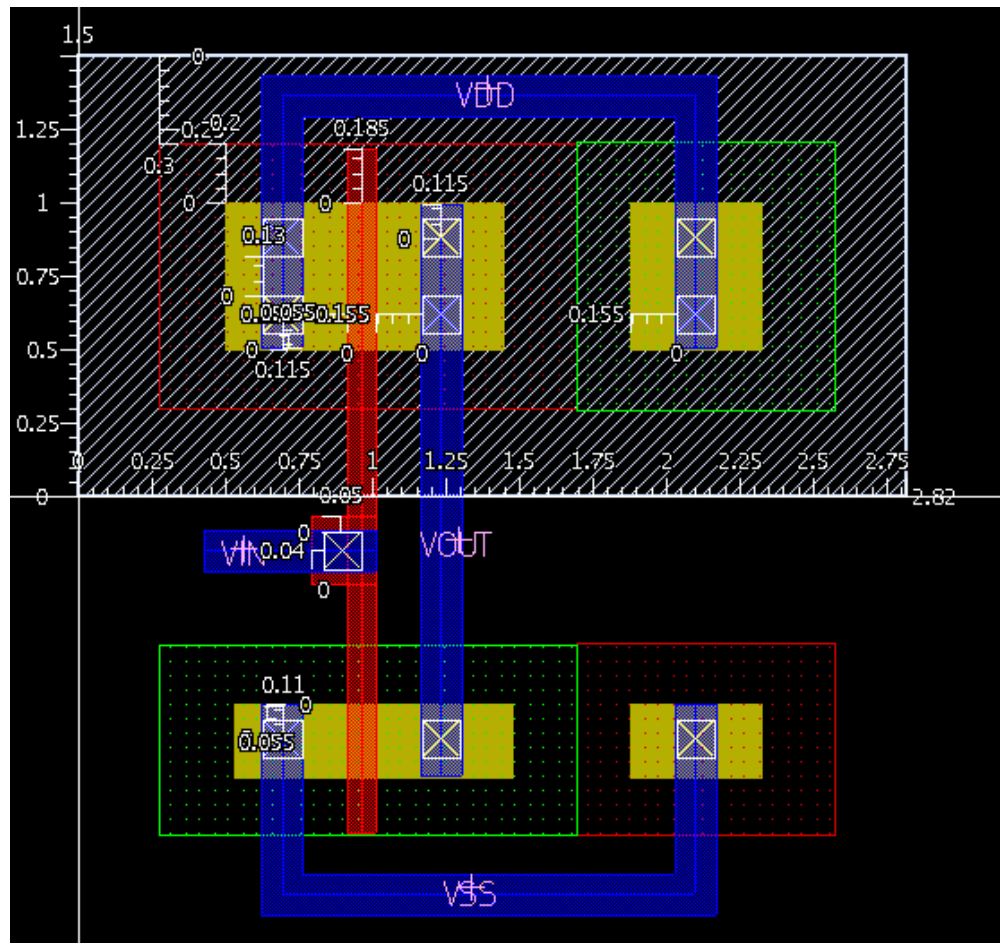


Figure 1: Inverter Layout

1.2 DRC Result (CLEAN)

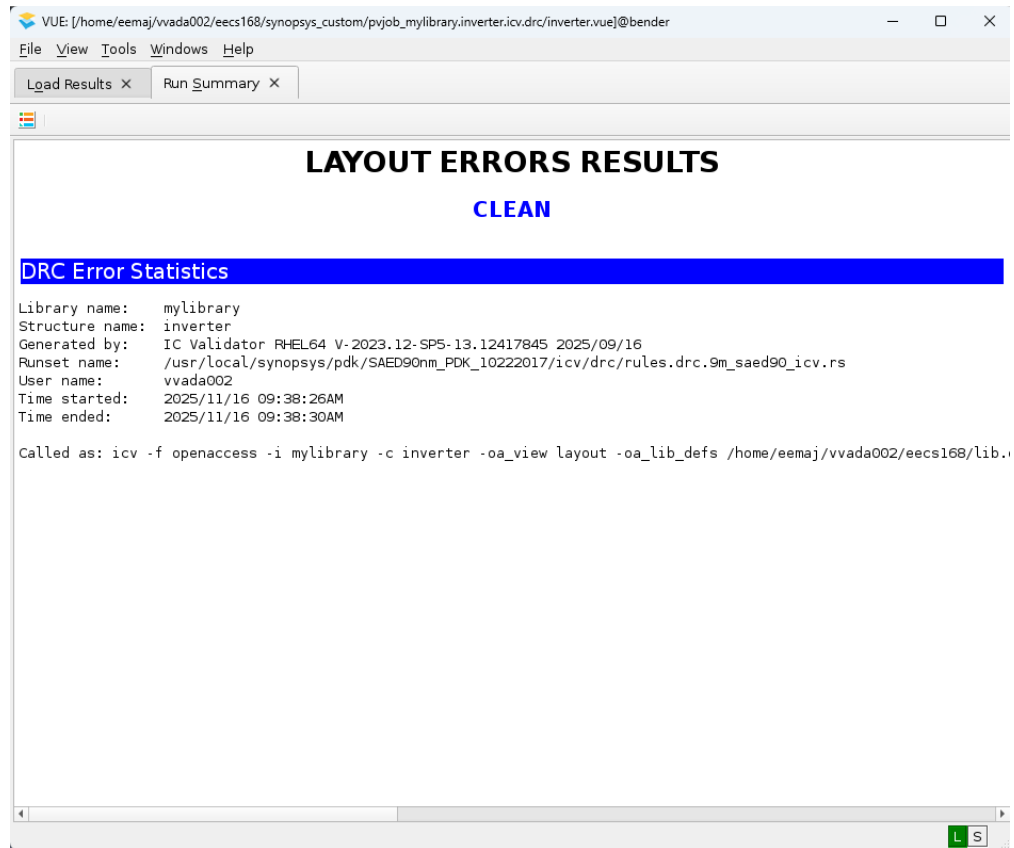


Figure 2: DRC: CLEAN

2 NAND Gate: Schematic, Simulation, Layout

2.1 NAND Schematic

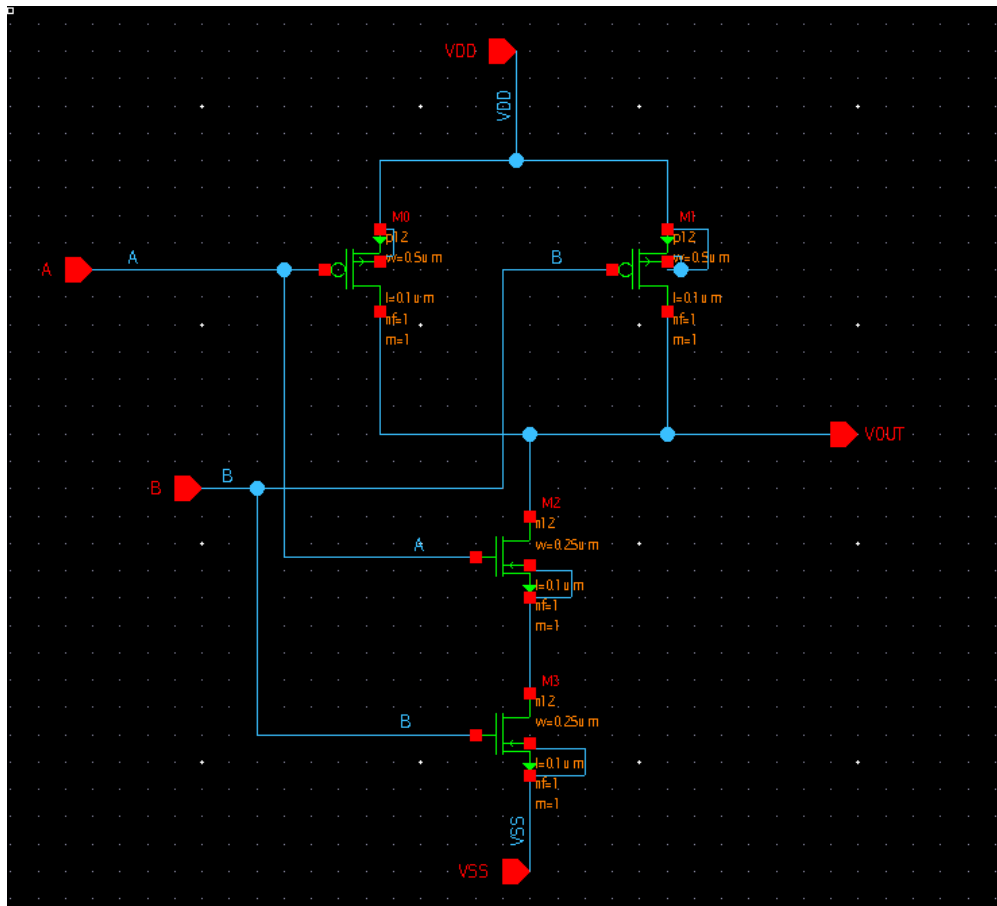


Figure 3: NAND Gate Schematic

2.2 NAND Testbench

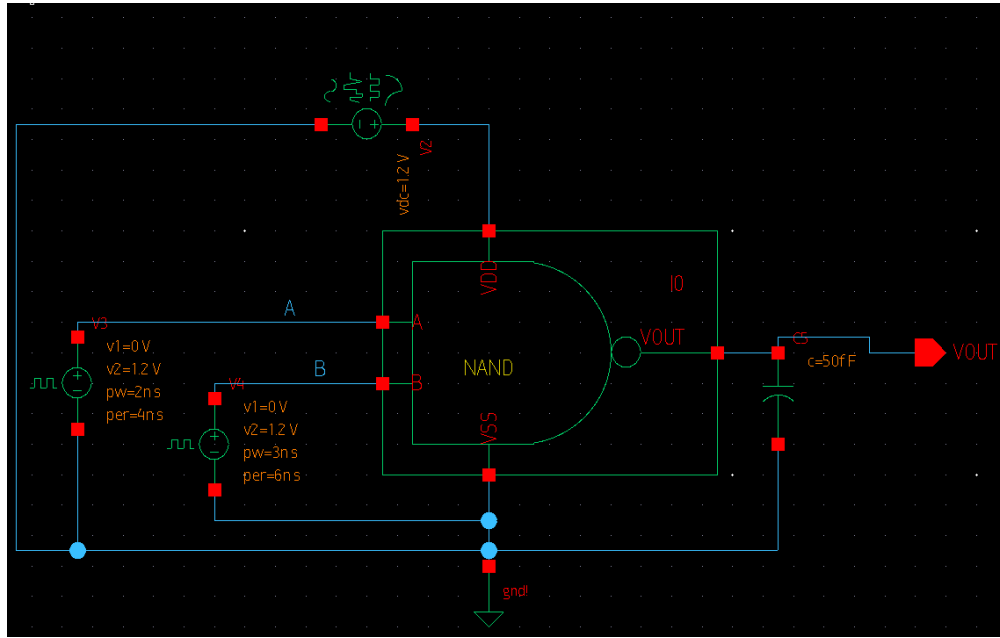


Figure 4: NAND Testbench Schematic

2.3 NAND DC Simulation Waveform

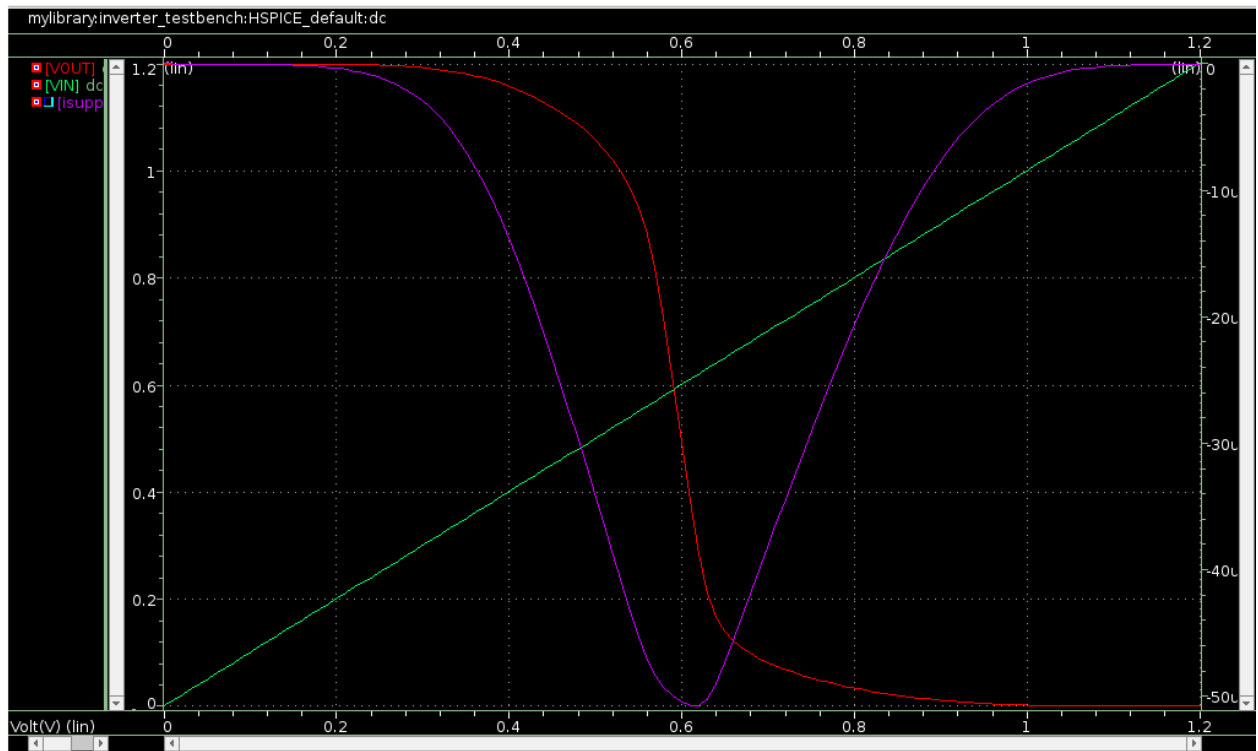


Figure 5: DC Transient Simulation Waveform for NAND Gate

2.4 NAND Trans Simulation Waveform

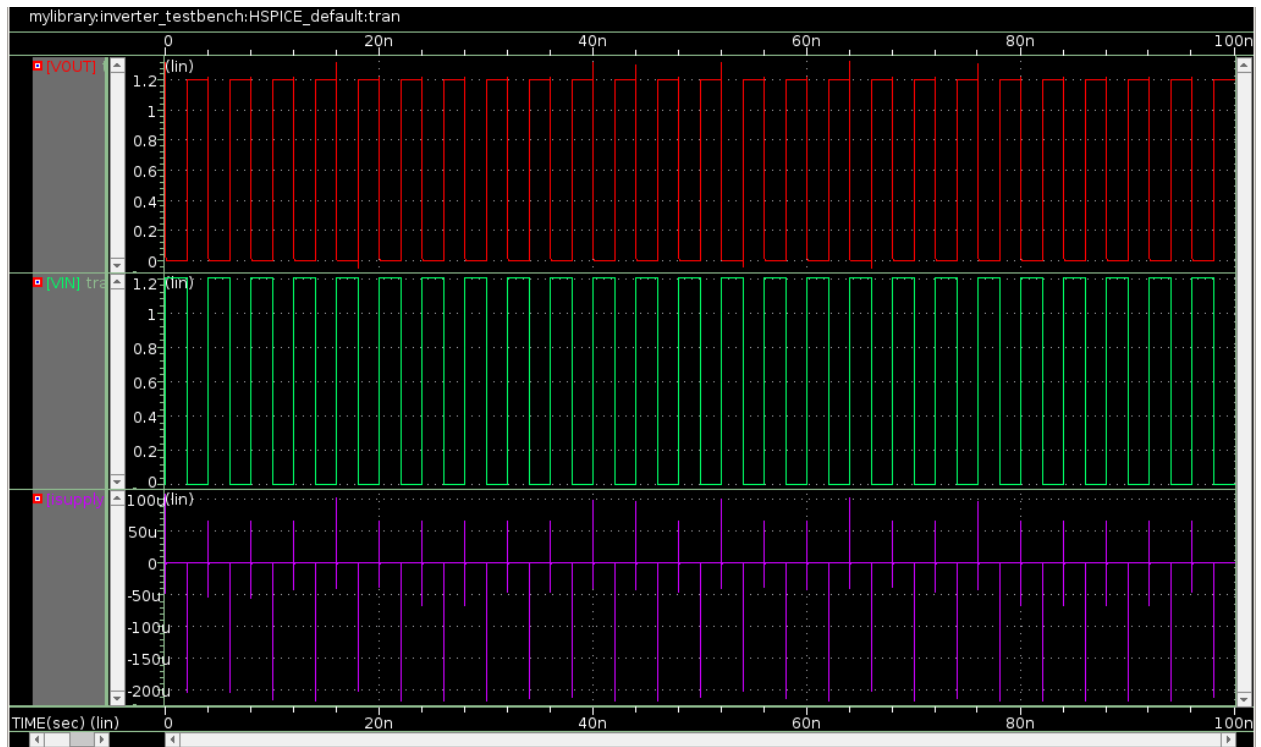


Figure 6: TRANS Transient Simulation Waveform for NAND Gate

2.5 NAND Layout

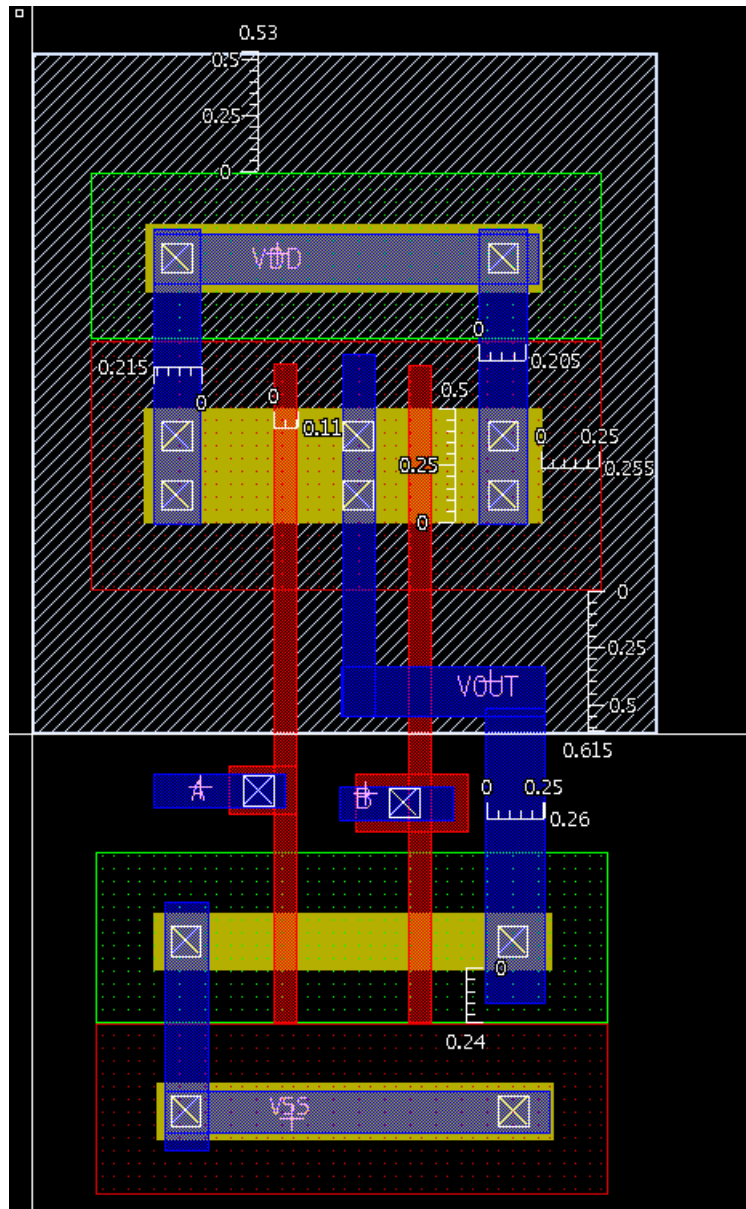


Figure 7: NAND Gate Layout

2.6 DRC Result (CLEAN)

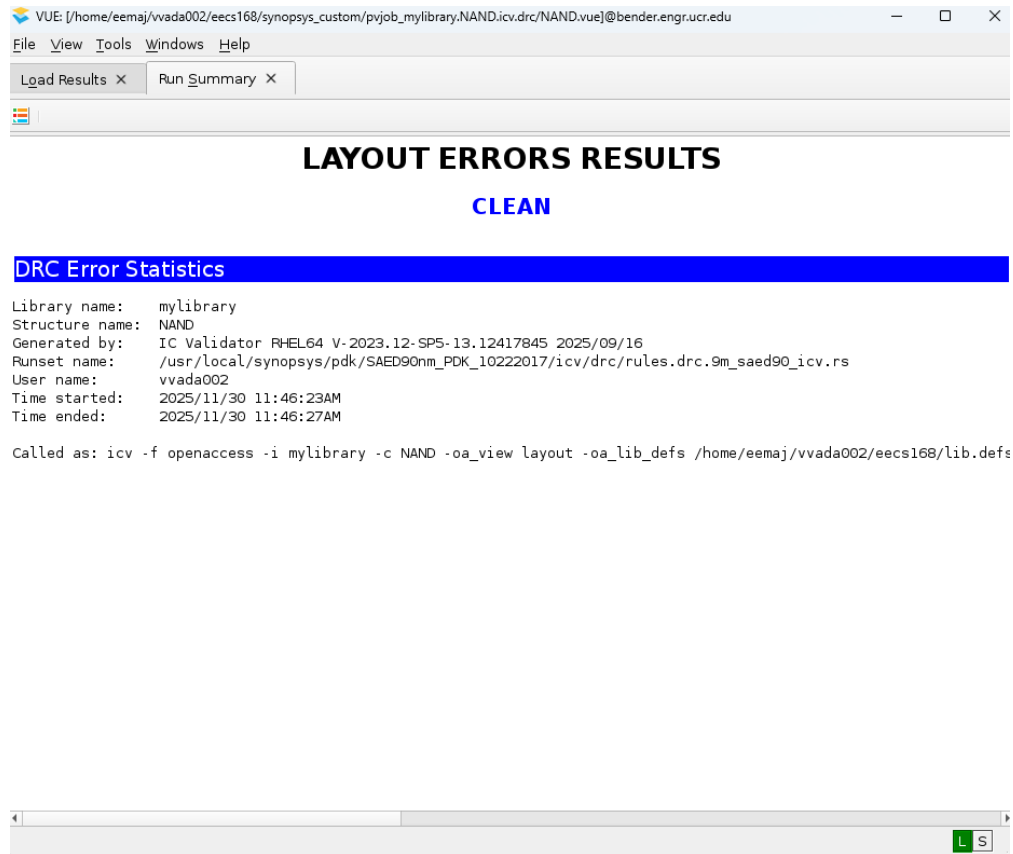


Figure 8: DRC: CLEAN (NAND)

3 Ring Oscillator (Hierarchical Design)

3.1 Ring Oscillator Schematic

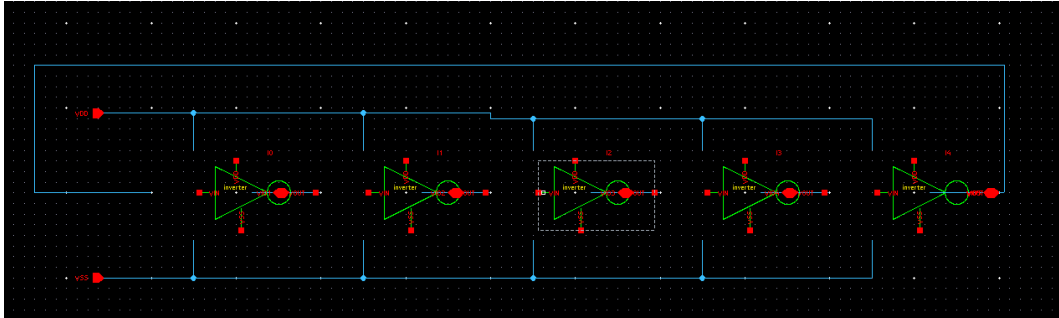


Figure 9: Ring Oscillator Schematic

3.2 Ring Oscillator Layout



Figure 10: Ring Oscillator Layout

3.3 Ring Oscillator Testbench

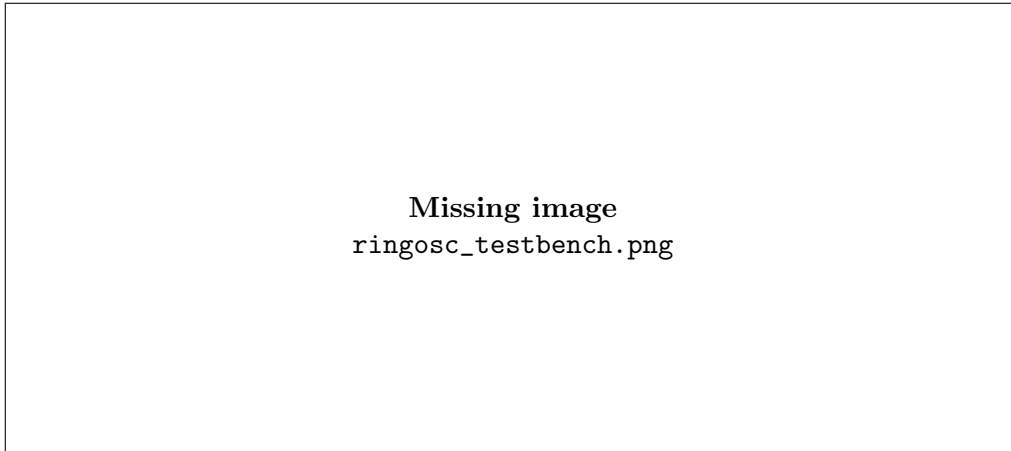


Figure 11: Ring Oscillator Testbench

3.4 DRC Result (CLEAN)



Figure 12: DRC: CLEAN (Ring Oscillator)

4 Issues Encountered

Write about any problems you encountered here.